Enantioselective Copper-Catalyzed Conjugate Addition of Dimethylzinc to 5-(1-Arylalkylidene) Meldrum's Acids

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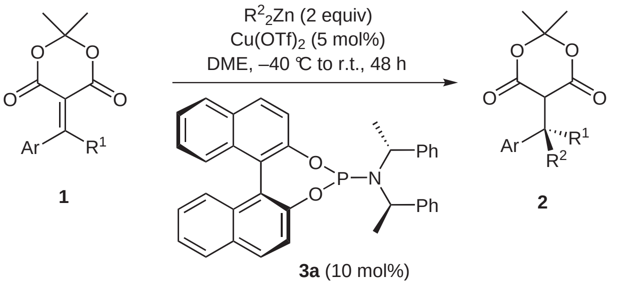
Abstract: The asymmetric formation of all-carbon quaternary benzylic stereocenters containing a methyl substituent was realized by the enantioselective copper-catalyzed 1,4-addition of dimethylzinc to alkylidene Meldrum's acids in the presence of a chiral phosphoramidite. Copper source and ligand optimization studies are presented, as well as the scope and limitations of dimethylzinc addition to various alkylidene Meldrum's acids. Good to excellent yields and enantioselectivities were obtained.

Key words: Meldrum's acid, all-carbon quaternary stereocenters, enantioselective conjugate addition, copper catalysis, organozinc reagents, phosphoramidite ligand

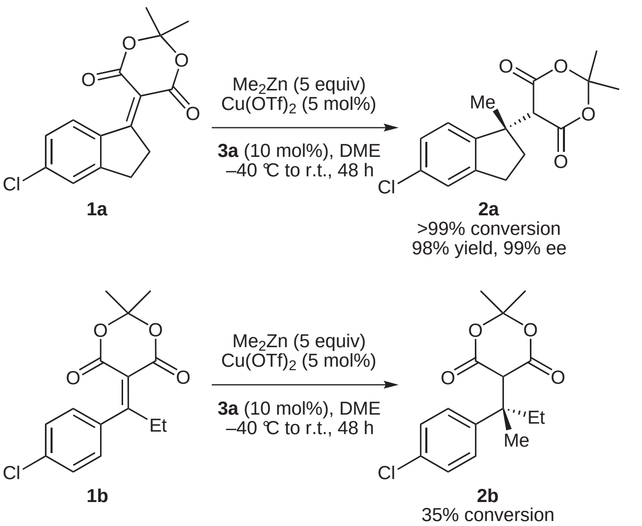
The catalytic asymmetric formation of all-carbon quaternary stereocenters remains an important goal in organic chemistry.[1] Conjugate addition of organometallic reagents to carbonyl-, nitro-, and sulfone-activated tri- and tetrasubstituted alkenes, has become a general and efficient method to access this structural motif.[2] A number of research groups,[3] along with ours,[4] have reported the enantioselective copper-catalyzed addition of organoaluminum, organozinc, and Grignard reagents. The rhodium-catalyzed addition of alkenyl- and arylboronic acids was also described.[5] From these synthetic strategies towards the formation of all-carbon quaternary stereocenters, aryl, alkenyl, and alkyl groups were introduced.

Our group[4] has established that alkylidene Meldrum's acids **1** are excellent electrophiles[6] to access all-carbon benzylic quaternary stereocenters through enantioselective copper-catalyzed conjugate addition of organozinc reagents in the presence of commercially available chiral phosphoramidite[7] ligand **3a**. Excellent conversions, yields, and enantioselectivities in the formation of **2** were achieved (Scheme 1). We also demonstrated that the pattern of substitution of the arene moiety impacted the efficiency of the reaction as well as the enantioselectivity of the conjugate addition.

Despite the superior electrophilicity of alkylidene Meldrum's acids **1** towards organozinc reagents, the addition of dimethylzinc lacked generality. The conjugate addition of dimethylzinc to Meldrum's acid indelydine **1a** gave >99% conversion, 98% isolated yield, and 99% ee.[4c] However, less reactive alkylidene **1b** led to a poor 35% conversion (Scheme 2).[4c]



Scheme 1 Enantioselective copper-catalyzed 1,4-addition of dialkylzinc reagents to alkylidene Meldrum's acids **1**



Scheme 2 Dimethylzinc addition to alkylidene Meldrum's acids

The commonality of methyl-containing all-carbon quaternary stereocenters in bioactive natural products prompted us to optimize the enantioselective methyl delivery to alkylidene Meldrum's acids **1**. A limited number of reports have appeared in the literature regarding asymmetric addition of methyl groups for the formation of all-carbon quaternary carbons. Hoveyda showed that the poorly nucleophilic dimethylzinc reacted with highly activated alkenes such as nitroalkenes,[3n] and cycloalkenones bearing a carboxylate group at the 2- or 3-position,[3m,k] in the presence of a copper catalyst, to yield all-carbon quaternary stereocenters in good yields and enantioselectivities. Our group obtained limited success on the addition of dimethylzinc to 2-(2,2-dimethyl-4,6-dioxo-1,3-dioxan-5-ylidene)aryl acetates.[4a] The addition of methylmagnesium bromide catalyzed by copper salts was also reported.[3a,f,o]

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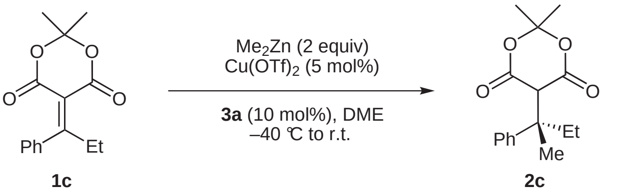
To date, the copper-catalyzed addition of trimethylaluminum to cyclic enones has provided the best selectivity, as independently reported by the Alexakis[3c–e,h] and Hoveyda[3i,j] groups. In this report, an optimized 1,4-conjugate addition of dimethylzinc to tetrasubstituted alkenes **1** is outlined along with substrate scope and limitations.

The asymmetric conjugate addition was optimized using 2,2-dimethyl-5-(1-phenylpropylidene)-1,3-dioxane-4,6-dione (**1c**) (Table 1). Subjecting **1c** under our previously developed conditions, 2 equivalents of dimethylzinc (10 wt% solution in hexane) and 5 mol% of copper(II) triflate in DME (final concentration of 0.08 M) from –40 °C to room temperature, gave a 32% conversion to Meldrum's acid **2c** after 48 hours (Table 1, entry 1). Strongly Lewis acidic nucleophiles were then investigated to promote the addition to the alkylidenes with good conversions. Grignard reagents methylmagnesium iodide, methylmagnesium bromide, and methylmagnesium chloride gave excellent conversions but nearly racemic product **2c** (Table 1, entries 2–4). The use of the (*R*,*S*)-Josiphos ligand also gave excellent conversions with nearly racemic products (not shown in Table 1). Conjugate addition involving organoaluminum reagents led to improved conversions in comparison to dimethylzinc. However, poor enantioselectivities were observed (entries 5 and 6). The nearly racemic product **2c** observed for these reactions suggested that the uncatalyzed addition of the nucleophiles towards alkylidene Meldrum's acid **1c** was the predominant pathway.

Table 1 Survey of Organometallic Reagents in the Conjugate Addition to Alkylidene Meldrum's Acid **1c**

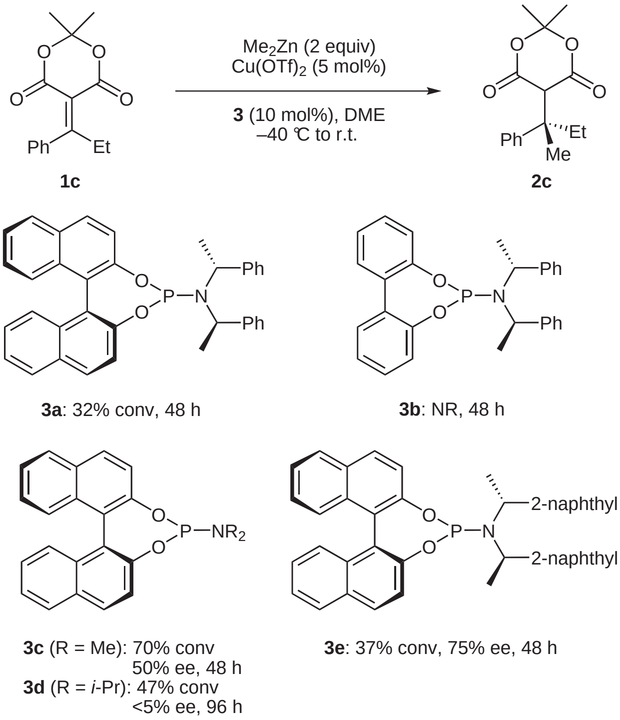


Entry	Me_nM	Time (h)	Conversion (%)	ee (%)
1	Me_2Zn	48	32	n/a[a]
2	MeMgI	72	63	<5
3	MeMgBr	72	88	<5
4	MeMgCl	72	97	<5
5	Me_2AlCl	96	64	<5
6	Me_3Al	48	96	<5

[a] Not applicable.

The tendancy of Grignard reagents and trimethylaluminum to give excellent conversion but racemic material led us to reexamine the asymmetric conjugate addition of dimethylzinc. We decided to systematically study the phosphoramidite ligand and copper sources to improve conversions. Ligand **3b**, containing a biphenyl rather than an axially chiral BINOL moiety as found in ligand **3a**, led to complete inhibition of the conjugate addition (Scheme 3). In contrast, leaving the BINOL moiety intact while replacing the chiral amine on the phosphorus atom with a dimethylamino group (**3c**) gave 70% conversion and 50% ee after 48 hours. Increasing the steric hindrance around the amine moiety by having a diisopropylamine (**3d**) had a detrimental effect on both conversion and enantioselectivity. The substitution of the phenyl moiety on the chiral amine with the 2-napthyl moiety **3e** gave only slight improvement of conversion with an enantioselectivity of 75% ee.

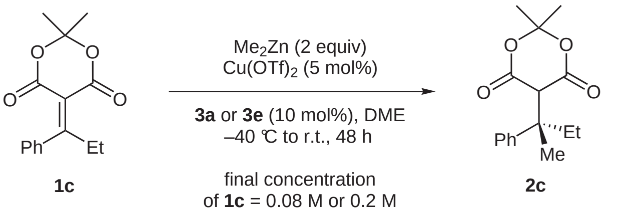
From this survey of phosphoramidite ligands, decreasing the size of the amine moiety increased the conversion, while decreasing enantioselectivity. These results showed that phosphoramidite ligand **3a** and 2-naphthyl substituted ligand **3e** led to optimal enantioselectivities despite lower conversions.



Scheme 3 Phosphoramidite ligands

To promote the addition of dimethylzinc to alkylidene Meldrum's acids **1**, a bimolecular process, the concentration of the reaction mixture was increased. Two approaches were used to run the reaction at higher concentration; either utilizing a more concentrated solution of dimethylzinc, or diminishing the quantity of DME. The reactions discussed above were carried out at a final concentration of 0.08 M for **1c**, considering for the total volume of solvent DME and hexanes from the 10 wt% solution of dimethylzinc. Gratifying, conversion improved substantially by decreasing the amount of DME used in half, for a final concentration of 0.2 M. As depicted in Table 2, when ligand **3a** was used, conversion increased from 32% to 78%, and 37% to 82% with ligand **3e**. Of

Table 2 Concentration Studies

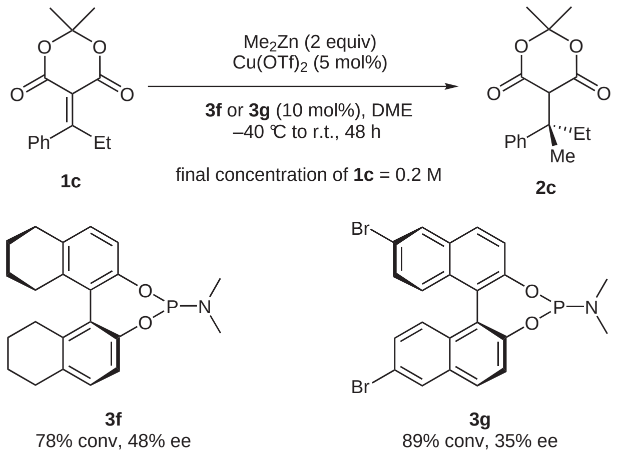


Entry	Ligand	Final concentration of **1c**	Conversion (%)	ee (%)
1	**3a**	0.08 M	32	n/a[a]
2	**3a**	0.2 M	78	77
3	**3e**	0.08 M	37	75
4	**3e**	0.2 M	82	75

[a] Not applicable.

note, the decrease in the relative quantity of DME versus hexane did not result in a loss in enantioselectivity.

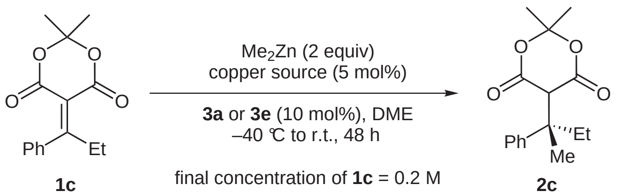
Further ligands with modifications to the BINOL backbone were tested to increase the conversion and the enantioselectivity (Scheme 4). Ligand **3f**, with a hydrogenated BINOL moiety, led to enantioselectivity similar to **3c**. Finally, 6,6′-dibrominated phosphoramidite ligand **3g** gave improved conversion but decreased enantioselectivity in comparison to **3c**. Based on these results, phosphoramidite **3a** and **3e** were the best ligands in the dimethylzinc addition for obtaining high yield and enantioselectivity.



Scheme 4 Varying the phosphoramidite ligand

Copper sources were then explored to further optimize conversion and enantioselectivity.[8] After testing a few copper sources, it was shown that consistent enantioselectivities were achieved regardless of the copper source for both ligands **3a** (Table 3, entries 1–5) and **3e** (Table 3, entries 6–8). However, the source of copper had a bigger influence on conversion. Copper(I) triflate gave the lowest conversion compared to all other copper sources tested (Table 3, entries 1 and 6), while $Cu(O_2CCF_3)_2{\cdot}\times H_2O$ led to the best results (entry 7). This time, the final concentration was increased, by using a more concentrated solution of dimethylzinc. The conversion was increased to near completion by using a 2.0 M solution of dimethylzinc in toluene (2 equiv) (Table 1, entry 8), for a final concentration of 0.33 M.

Table 3 Optimization of the Copper Source



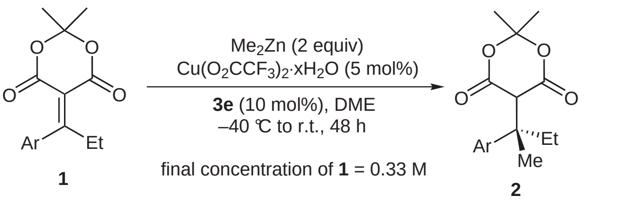
Entry	Ligand	Copper source	Conversion (%)	ee (%)
1	**3a**	$Cu(OTf)_2$	78	77
2	**3a**	$Cu(OAc)_2{\cdot}H_2O$	94	75
3	**3a**	$Cu(acac)_2$	88	75
4	**3a**	CuTC[a]	90	74
5	**3a**	$Cu(O_2CCF_3)_2{\cdot}\times H_2O$	87	73
6	**3e**	$Cu(OTf)_2$	82	75
7	**3e**	$Cu(O_2CCF_3)_2{\cdot}\times H_2O$	95	77
8	**3e**	$Cu(O_2CCF_3)_2{\cdot}\times H_2O$	98[b]	76

[a] CuTC: copper(I) thiophene-2-carboxylate.
[b] Me_2Zn (2.0 M in toluene) was used rather than Me_2Zn (10 wt% in hexane). Final concentration of **1c** was 0.33 M.

With the optimal conditions in hand, the scope of the dimethylzinc addition to alkylidene Meldrum's acids was investigated (Table 4).[9] Electron-donating groups positioned *para* to the electrophilic site furnished products in excellent yields and enantioselectivities (Table 4, entries 2–4, 7). Electron-withdrawing groups at the 4-position also provided products with good yields and enantioselectivities (Table 4, entries 5, 6). Substituting the *meta* position yielded diminished selectivities, while *ortho*-substitution led to fully recovery of starting material. These results are consistent with our previously reported diethylzinc addition to similar alkylidenes.[4c] Interestingly, 3-*N*-tosylpyrrole-substituted **1l** gives product **2l** in excellent yield and enantioselectivity (Table 1, entry 11).

It is noteworthy to comment on the experimental procedure. A typical reaction involves the addition of dimethylzinc to a precooled solution containing copper source and the chiral phosphoramidite ligand. After five minutes, a solution of the substrate in DME is added to the reaction mixture. Although, this procedure works with a number of these alkylidenes, it became a problem for some substrates insoluble in DME at high concentration. To overcome this problem, a solution of copper, phosphoramidite ligand, and dimethylzinc was added onto a concentrated mixture of the substrate in DME. This 'reverse' addition

Table 4 Scope of the Conjugate Addition of Dimethylzinc



Entry	Ar	Product **2**	Conversion (%)	Yield (%)	ee (%)
1	Ph (**1c**)	**2c**	98	93	76
2	4-MeC_6H_4 (**1d**)	**2d**	99	85	83
3	4-*i*-PrC_6H_4 (**1e**)	**2e**	99	99	94
4	4-*t*-BuC_6H_4 (**1f**)	**2f**	99	84	97
5	4-BrC_6H_4 (**1g**)	**2g**	98	68	91
6	4-ClC_6H_4 (**1b**)	**2b**	96	68	90
7	4-$MeOC_6H_4$ (**1h**)	**2h**	99	93	85
8	3-$MeOC_6H_4$ (**1i**)	**2i**	97	79	69
9	2-$MeOC_6H_4$ (**1j**)	**2j**	0	n/a	n/a
10	2-naphthyl (**1k**)	**2k**	99	82	79
11	3-(*N*-Ts)pyrrolyl (**1l**)	**2l**	>99	93	97

[a] Not applicable.

allowed for better conversion and isolated yields while not affecting the enantioselectivities.[3d]

In summary, the asymmetric conjugate addition of dimethylzinc to alkylidene Meldrum's acids has been achieved in good to excellent yields and enantioselectivities. The application of this methodology to natural product synthesis is in progress.

All reactions were carried out in flame-dried glassware under a dry argon atmosphere. 1,2-Dimethoxyethane (DME) was distilled from sodium-benzophenone ketyl under N_2 and degassed via freeze-pump-thaw method. All copper sources were obtained from commercial sources and used without further purification. Me_2Zn (1.0 M in heptanes), and Me_2Zn (2.0 M in toluene), were also obtained from commercial sources and used without further purification. Chiral phosphoramidite ligands were either obtained from commercial sources or prepared according to literature procedures.[10] ^{1}H NMR spectra were referenced to residual ^{1}H shift in $CDCl_3$ (7.24 ppm). $CDCl_3$ (77.0 ppm) was used as the internal reference for ^{13}C NMR spectra. Flash chromatography was performed using 230–400 mesh silica gel. Melting points are uncorrected. Optical rotations were recorded in cells with 1 dm path length. Chiral HPLC analyses were performed using a Chiralcel OD-H or AD-H column. All columns are 250 × 4.6 mm. High-resolution mass spectra were run at the University of Waterloo with a source temperature of 200 °C, mass resolution of 9000, and electron energy of 70 eV.

Alkylidene Meldrum's Acids 1; General Procedure A

Alkylidene Meldrum's acids were prepared by the Knoevenagel condensation of Meldrum's acid with ketones using Brown and coworkers' method.[11] In a typical reaction, a solution of $TiCl_4$ (3.5 mL, 32 mmol, 2.1 equiv) in CH_2Cl_2 (6 mL) was added dropwise under N_2 to anhyd THF (55 mL), which was cooled at 0 °C, resulting in a yellow suspension. A solution containing the ketone (15 mmol, 1.0 equiv) and Meldrum's acid (15 mmol, 1.0 equiv) in anhyd THF (10 mL) was added dropwise via a syringe to the $TiCl_4$·THF complex. The flask containing the solution of ketone and Meldrum's acid was rinsed with THF (10 mL) and added to the mixture. Subsequently, pyridine (75 mmol, 5.0 equiv) was added to the reaction mixture dropwise at 0 °C. The reaction was allowed to warm up slowly to r.t. and stirred until completion of reaction or for 24 h. The reaction was quenched by the addition of H_2O (100 mL) and diluted with either Et_2O or EtOAc (100 mL). After the solid had dissolved, the layers were partitioned. The aqueous layer was extracted with Et_2O, or EtOAc (2 × 100 mL), and the combined organic layers were washed with aq sat. $NaHCO_3$ (2 × 100 mL), brine (50 mL), dried ($MgSO_4$), filtered, and concentrated. Purification by either crystallization or trituration with MeOH provided the alkylidene Meldrum's acids. For data on **1a**, see ref. 4c.

5-[1-(4-Chlorophenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1b)

Yield: 15%; white solid; mp 120–122 °C (MeOH).

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.36 (d, *J* = 8.4 Hz, 1 H), 7.08 (d, *J* = 8.4 Hz, 1 H), 3.07 (q, *J* = 7.5 Hz, 2 H), 1.78 (s, 6 H), 1.05 (t, *J* = 7.5 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ =176.6, 160.7, 160.1, 138.2, 135.2, 128.6, 127.6, 117.2, 103.9, 31.4, 27.3, 12.3.

HRMS (EI): *m*/*z* calcd for $C_{15}H_{15}ClO_4$ (M^+): 294.0659; found: 294.0668.

2,2-Dimethyl-5-(1-phenylpropylidene)-1,3-dioxane-4,6-dione (1c)

Yield: 43%; white solid; mp 98–100 °C (MeOH).

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.40–7.38 (m, 3 H), 7.15–7.13 (m, 2 H), 3.12 (q, *J* = 7.4 Hz, 2 H), 1.80 (s, 6 H), 1.06 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 178.0, 160.9, 160.3, 140.0, 129.1, 128.3, 126.1, 116.8, 103.8, 31.3, 27.2, 12.3.

HRMS (EI): *m*/*z* calcd for $C_{15}H_{16}O_4$ (M^+): 260.1049; found: 260.1040.

2,2-Dimethyl-5-(1-*p*-tolylpropylidene)-1,3-dioxane-4,6-dione (1d)

Yield: 7% (unoptimized); white solid; mp 145–147 °C (MeOH).

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.20 (d, *J* = 7.6 Hz, 2 H), 7.06 (d, *J* = 7.7 Hz, 2 H), 3.12 (q, *J* = 7.4 Hz, 2 H), 2.37 (s, 3 H), 1.80 (s, 6 H), 1.05 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 178.4, 161.0, 160.6, 139.5, 137.1, 129.1, 126.3, 116.5, 103.7, 31.3, 27.3, 21.4, 12.5.

HRMS (EI): *m*/*z* calcd for $C_{16}H_{18}O_4$ (M^+): 274.1205; found: 274.1201.

5-[1-(4-Isopropylphenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1e)

Yield: 33%; light yellow solid; mp 106–108 °C (MeOH).

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.24 (d, *J* = 7.5 Hz, 2 H), 7.10 (d, *J* = 7.6 Hz, 2 H), 3.11 (q, *J* = 7.3 Hz, 2 H), 2.91 (sept, *J* = 6.8 Hz, 1 H), 1.80 (s, 6 H), 1.24 (d, *J* = 6.9 Hz, 2 H), 1.05 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 178.2, 160.9, 160.5, 150.1, 137.2, 126.4, 126.3, 116.3, 103.6, 33.7, 31.2, 27.1, 23.6, 12.5.

HRMS (EI): *m*/*z* calcd for $C_{18}H_{22}O_4$ (M^+): 302.1558; found: 302.1521.

5-[1-(4-*tert*-Butylphenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1f)
Yield: 26%; white solid; mp 110–112 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.39 (d, *J* = 7.9 Hz, 2 H), 7.11 (d, *J* = 7.9 Hz, 2 H), 3.11 (q, *J* = 7.4 Hz, 2 H), 1.80 (s, 6 H), 1.31 (s, 9 H), 1.06 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 178.4, 161.0, 160.7, 152.5, 137.0, 126.2, 125.2, 116.4, 103.7, 34.7, 31.3, 31.1, 27.2, 12.6.

HRMS (EI): *m/z* calcd for $C_{16}H_{18}O_5$ (M^+): 316.1675; found: 316.1673.

5-[1-(4-Bromophenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1g)
Yield: 25%; white solid; mp 124–125 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.51 (d, *J* = 8.1 Hz, 2 H), 7.01 (d, *J* = 8.1 Hz, 2 H), 3.06 (q, *J* = 7.4 Hz, 2 H), 1.77 (s, 6 H), 1.03 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 176.5, 160.6, 160.1, 138.7, 131.4, 127.7, 123.3, 117.0, 103.9, 31.2, 27.2, 12.2.

HRMS (EI): *m/z* calcd for $C_{15}H_{15}BrO_4$ (M^+): 338.0154; found: 338.0160.

5-[1-(4-Methoxyphenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1h)
Yield: 29%; light yellow solid; mp 113–114 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.16 (d, *J* = 7.8 Hz, 2 H), 6.91 (d, *J* = 7.8 Hz, 2 H), 3.82 (s, 3 H), 3.13 (q, *J* = 7.3 Hz, 2 H), 1.80 (s, 6 H), 1.05 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 178.2, 161.1, 161.0, 160.9, 132.0, 128.6, 115.8, 113.8, 103.6, 55.3, 31.3, 27.3, 12.9.

HRMS (EI): *m/z* calcd for $C_{16}H_{18}O_5$ (M^+): 290.1154; found: 290.1156.

5-[1-(3-Methoxyphenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1i)
Yield: 26%; light yellow solid; mp 72–74 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.32 (t, *J* = 7.9 Hz, 1 H), 6.91 (dd, *J* = 8.3, 2.4 Hz, 1 H), 6.71–6.67 (m, 2 H), 3.80 (s, 3 H), 3.10 (q, *J* = 7.5 Hz, 2 H), 1.79 (s, 3 H), 1.06 (t, *J* = 7.5 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 177.5, 160.9, 160.2, 159.4, 141.4, 129.5, 118.4, 117.0, 113.8, 112.3, 103.8, 55.2, 31.2, 27.3, 12.3.

HRMS (EI): *m/z* calcd for $C_{16}H_{18}O_5$ (M^+): 290.1154; found: 290.1159.

5-[1-(2-Methoxyphenyl)propylidene]-2,2-dimethyl-1,3-dioxane-4,6-dione (1j)
Yield: 64%; light yellow solid; mp 105–107 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.33 (dt, *J* = 7.7, 1.8 Hz, 1 H), 7.08–6.99 (m, 2 H), 6.88 (d, *J* = 8.3 Hz, 1 H), 3.74 (s, 3 H), 3.36 (dq, *J* = 12.9, 7.4 Hz, 1 H), 2.85 (dq, *J* = 12.8, 7.6 Hz, 1 H), 1.86 (s, 3 H), 1.73 (s, 3 H), 1.02 (t, *J* = 7.5 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 174.0, 161.2, 160.4, 154.2, 130.0, 129.0, 127.3, 120.7, 118.0, 110.8, 103.9, 55.2, 30.7, 27.6, 26.4, 11.8.

HRMS (EI): *m/z* calcd for $C_{16}H_{18}O_5$ (M^+): 290.1154; found: 290.1163.

2,2-Dimethyl-5-[1-(naphthalen-2-yl)propylidene]-1,3-dioxane-4,6-dione (1k)
Yield: 14%; light yellow solid; mp 123–125 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.87–7.82 (m, 3 H), 7.63 (s, 1 H), 7.52–7.49 (m, 2 H), 7.26 (d, *J* = 8.6 Hz, 1 H), 3.23 (q, *J* = 7.5 Hz, 2 H), 1.84 (s, 6 H), 1.09 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 178.2, 161.0, 160.4, 137.5, 133.3, 132.7, 128.3, 128.0, 127.8, 126.9, 126.6, 125.3, 124.3, 116.9, 103.8, 31.6, 27.3, 12.5.

HRMS (EI): *m/z* calcd for $C_{19}H_{18}O_4$ (M^+): 310.1205; found: 310.1213.

2,2-Dimethyl-5-[1-(1-tosyl-1*H*-pyrrol-3-yl)propylidene]-1,3-dioxane-4,6-dione (1l)
Yield: 39%; light yellow solid; mp 133–134 °C (MeOH).

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.77 (d, *J* = 8.3 Hz, 2 H), 7.64 (s, 1 H), 7.31 (d, *J* = 8.1 Hz, 2 H), 7.11–7.09 (m, 1 H), 6.39–6.37 (m, 1 H), 2.99 (q, *J* = 7.4 Hz, 2 H), 2.39 (s, 3 H), 1.73 (s, 6 H), 1.12 (t, *J* = 7.4 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 168.2, 161.2, 161.1, 145.8, 135.2, 130.3, 127.1, 125.6, 123.8, 120.9, 114.6, 114.0, 103.6, 30.8, 27.1, 21.7, 14.1.

HRMS (EI): *m/z* calcd for $C_{20}H_{21}NO_6S$ (M^+): 403.1090; found: 403.1082.

Asymmetric 1,4-Conjugate Addition of Me_2Zn to Alkylidene Meldrum's Acids 1; General Procedure B
Reactions were typically carried out using 0.4 mmol of substrate. In a glove box, copper source (5 mol%) and phosphoramidite chiral ligand (10 mol%) were added to a flame-dried resealable Schlenk tube. DME (0.3 mL) was then added to the Schlenk tube to wash down any residual solids to the bottom. The reaction mixture was allowed to stir at r.t. for 30 min, outside the glove box, and then cooled to –40 °C. In the dry box, Me_2Zn solution (2.0 equiv) was transferred to a round-bottom flask equipped with a septum. This solution was added to the Schlenk tube dropwise via a syringe and the resulting solution stirred for 5 min. A solution of alkylidene Meldrum's acid **1** (1.0 equiv) in DME (0.3 mL) was then added dropwise via a syringe. Finally, DME (0.2 mL) was added to wash down the remaining solid on the sides of the Schlenk tube. The mixture was allowed to warm up slowly to r.t. After stirring for 48 h, the mixture was cooled to 0 °C, aq 5% HCl (5 mL) and EtOAc (20 mL) were added to the mixture. The layers were partitioned and the aqueous layer was extracted with EtOAc (3 × 20mL). The combined organic layers were washed with brine (20 mL), dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by flash column chromatography on silica gel using hexanes in EtOAc as eluent to yield the desired product. HPLC using a chiral column (OD-H or AD-H) was used to measure the enantiomeric ratio of the products. For data on **2a**, see ref. 4c.

(*S*)-5-[1-(4-Chlorophenyl)-1-methylpropyl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2b)
Yield: 68%; white solid; mp 92–97 °C; $[\alpha]_D^{30}$ –12.1 (*c* 2.8, CH_2Cl_2).

An enantiomeric excess of 90% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 20.4 min (*S*), t_{R2} = 25.6 min (*R*)].

^{1}H NMR (CDCl$_3$, 300 MHz): δ = 7.29 (d, *J* = 8.7 Hz, 2 H), 7.20 (d, *J* = 8.7 Hz, 2 H), 3.60 (s, 1 H), 2.10 (q, *J* = 7.3 Hz, 2 H), 1.63 (s, 3 H), 1.58 (s, 3 H), 1.33 (s, 3 H), 0.72 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR (CDCl$_3$, 75 MHz): δ = 164.3, 163.8, 141.2, 132.8, 128.4, 128.3, 105.1, 57.0, 45.7, 32.7, 29.1, 27.4, 21.4, 8.6.

HRMS (EI): *m/z* calcd for $C_{16}H_{19}ClO_4$ (M^+): 310.0972; found: 310.0975.

(*S*)-2,2-Dimethyl-5-(2-phenylbutan-2-yl)-1,3-dioxane-4,6-dione (2c)
Yield: 93%; white solid; mp 90–94 °C; $[\alpha]_D^{25}$ –0.32 (*c* 4.8, CH_2Cl_2).

An enantiomeric excess of 76% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 15.5 min (*S*), t_{R2} = 20.3 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.36–7.21 (m, 5 H), 3.64 (s, 1 H), 2.22 (dq, *J* = 7.2, 7.0, Hz 1 H), 2.11 (dq, *J* = 7.1, 7.0 Hz, 1 H), 1.64 (s, 3 H), 1.61 (s, 3 H), 1.19 (s, 3 H), 0.78 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.4, 163.9, 142.2, 128.2, 126.8, 126.7, 105.0, 57.0, 46.0, 32.5, 29.2, 27.0, 21.6, 8.6.

HRMS (EI): *m*/*z* calcd for $C_{16}H_{20}O_4$ (M^+): 276.1362; found: 276.1367.

(*S*)-2,2-Dimethyl-5-[1-methyl-1-(4-methylphenyl)propyl]-1,3-dioxane-4,6-dione (2d)
Yield: 85%; white solid; mp 90–94 °C; $[\alpha]_D^{26.5}$ +0.15 (*c* 4.4, CH_2Cl_2).

An enantiomeric excess of 83% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 15.3 min (*S*), t_{R2} = 19.1 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.17 (d, *J* = 13.8 Hz, 2 H), 7.12 (d, *J* = 13.8 Hz, 2 H), 3.55 (s, 1 H), 2.29 (s, 3 H), 2.17 (dq, *J* = 14.2, 7.2 Hz, 1 H), 2.06 (dq, *J* = 14.1, 7.2 Hz, 1 H), 1.58 (s, 6 H), 1.17 (s, 3 H), 0.75 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.7, 164.2, 139.1, 136.5, 129.0, 126.8, 105.2, 57.2, 46.1, 32.7, 29.5, 27.1, 22.0, 20.8, 8.7.

HRMS (EI): *m*/*z* calcd for $C_{17}H_{22}O_4$ (M^+): 290.1518; found: 290.1515.

(*S*)-5-[2-(4-*tert*-Butylphenyl)butan-2-yl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2f)
Yield: 84%; white solid; mp 100–103 °C; $[\alpha]_D^{31}$ +3.2 (*c* 2.8, CH_2Cl_2).

An enantiomeric excess of 97% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 11.7 min (*S*), t_{R2} = 14.3 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.32 (d, *J* = 8.6 Hz, 2 H), 7.20 (d, *J* = 8.6 Hz, 2 H), 3.49 (s, 1 H), 2.25 (dq, *J* = 14.1, 7.2 Hz, 1 H), 2.13 (dq, *J* = 14.1, 7.1 Hz, 1 H), 1.61 (s, 3 H), 1.54 (s, 3 H), 1.26 (s, 9 H), 0.96 (s, 3 H), 0.82 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.7, 164.3, 150.0, 138.6, 126.7, 125.3, 105.4, 57.1, 46.2, 34.3, 32.4, 31.2, 29.8, 26.7, 22.5, 8.8.

HRMS (EI): *m*/*z* calcd for $C_{20}H_{28}O_4$ (M^+): 332.1988; found: 332.1996.

(*S*)-5-[1-(4-Bromophenyl)-1-methylpropyl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2g)
Yield: 68%; white solid; mp 84–85 °C; $[\alpha]_D^{27.5}$ –10.8 (*c* 2.4, CH_2Cl_2).

An enantiomeric excess of 91% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 21.7 min (*S*), t_{R2} = 27.7 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.44 (d, *J* = 8.5 Hz, 2 H), 7.14 (d, *J* = 8.5 Hz, 2 H), 3.60 (s, 1 H), 2.10 (q, *J* = 7.3 Hz, 2 H), 1.63 (s, 3 H), 1.57 (s, 3 H), 1.34 (s, 3 H), 0.72 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.2, 163.8, 141.8, 131.4, 128.6, 120.9, 105.1, 56.9, 45.7, 32.7, 29.0, 27.5, 21.3, 8.6.

HRMS (EI): *m*/*z* calcd for $C_{16}H_{19}BrO_4$ (M^+): 354.0467; found: 354.0470.

(*S*)-5-[2-(4-Methoxyphenyl)butan-2-yl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2h)
Yield: 93%; clear oil; $[\alpha]_D^{23.5}$ +0.11 (*c* 5.5, CH_2Cl_2).

An enantiomeric excess of 85% *S* was measured by chiral HPLC [AD-H, 10% *i*-PrOH–hexanes, 1.0 mL/min, t_{R1} = 8.0 min (*S*), t_{R2} = 8.7 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.18 (d, *J* = 8.9 Hz, 2 H), 6.83 (d, *J* = 8.9 Hz, 2 H), 3.76 (s, 3 H), 3.50 (s, 1 H), 2.15 (dq, *J* = 14.2, 7.3 Hz, 1 H), 2.05 (dq, *J* = 14.1, 7.2 Hz, 1 H), 1.57 (s, 3 H), 1.56 (s, 3 H), 1.16 (s, 3 H), 0.76 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.7, 164.2, 158.4, 133.9, 128.1, 113.6, 105.2, 57.3, 55.2, 46.0, 32.7, 29.6, 27.1, 22.3, 8.7.

HRMS (EI): *m*/*z* calcd for $C_{17}H_{22}O_5$ (M^+): 306.1467; found: 306.1475.

(*S*)-5-[2-(3-Methoxyphenyl)butan-2-yl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2i)
Yield: 79%; clear oil; $[\alpha]_D^{25}$ –0.38 (*c* 4.8, CH_2Cl_2).

An enantiomeric excess of 69% *S* was measured by chiral HPLC [AD-H, 10% *i*-PrOH–hexanes, 1.0 mL/min, t_{R1} = 7.2 min (*S*), t_{R2} = 8.4 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.21 (t, *J* = 8.1 Hz, 1 H), 6.85 (d, *J* = 8.0 Hz, 1 H), 6.80 (br s, 1 H), 6.74 (dd, *J* = 8.1, 2.1 Hz, 1 H), 3.75 (s, 3 H), 3.60 (s, 1 H), 2.09 (dq, *J* = 18.5, 7.1 Hz, 2 H), 1.59 (s, 3 H), 1.57 (s, 3 H), 1.23 (s, 3 H), 0.73 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.4, 163.8, 159.5, 144.1, 129.1, 119.1, 113.4, 111.4, 105.0, 56.9, 55.1, 46.0, 32.7, 29.1, 27.2, 21.5, 8.6.

HRMS (EI): *m*/*z* calcd for $C_{17}H_{22}O_5$ (M^+): 306.1467; found: 306.1471.

'Reverse' Asymmetric 1,4-Conjugate Addition of Me_2Zn to Alkylidene Meldrum's Acids 1e,k,l; General Procedure C
Reactions were typically carried out using 0.4 mmol of substrate. In a glove box, copper source (5 mol%) and phosphoramidite chiral ligand (10 mol%) were added to a flame-dried flask equipped with a stir bar. DME (0.4 mL) was then added to the Schlenk tube to wash down any residual solids to the bottom. The reaction mixture was allowed to stir at r.t. for 30 min, outside the glove box, and then cooled to –40 °C. In the dry box, Me_2Zn solution (2.0 equiv) was transferred to a round-bottomed flask equipped with a septum. Me_2Zn solution was added to the mixture at –40 °C. After stirring for 5 min, the resulting mixture was added to a –40 °C, pre-cooled Schlenk tube charged with alkylidene Meldrum's acid **1e**,**k**,**l** (1.0 equiv) in DME (0.2 mL) dropwise via a syringe. DME (0.2 mL) was used to wash the flask and transfer remaining residue into the Schlenk tube. The mixture was allowed to warm up slowly to r.t. After stirring for 48 h, the mixture was cooled to 0 °C, 5% aq HCl (5 mL) and EtOAc (20 mL) were added to the mixture. The layers were partitioned and the aqueous layer was extracted with EtOAc (3 × 20 mL). The combined organic layers were washed with brine (20 mL), dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by flash column chromatography on silica gel using hexanes in EtOAc as eluent to yield the desired product. HPLC using a chiral column (OD-H or AD-H) was used to measure the enantiomeric ratio of the products.

(*S*)-5-[2-(4-Isopropylphenyl)butan-2-yl]-2,2-dimethyl-1,3-dioxane-4,6-dione (2e)
Yield: 99%; white solid; mp 80–82 °C; $[\alpha]_D^{29}$ +5.8 (*c* 4.8, CH_2Cl_2).

An enantiomeric excess of 94% *R* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 11.4 min (*S*), t_{R2} = 13.6 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.19 (d, *J* = 8.6 Hz, 2 H), 7.15 (d, *J* = 8.7 Hz, 2 H), 3.50 (s, 1 H), 2.85 (sept, *J* = 6.9 Hz, 1 H), 2.24 (dq, *J* = 14.3, 7.2 Hz, 1 H), 2.12 (dq, *J* = 14.1, 7.2 Hz, 1 H), 1.60 (s, 3 H), 1.54 (s, 3 H), 1.19 (d, *J* = 6.9 Hz, 6 H), 0.80 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.7, 164.2, 147.6, 139.0, 126.9, 126.4, 105.3, 57.1, 46.2, 33.5, 32.5, 29.7, 26.8, 23.8, 22.4, 8.8.

HRMS (EI): *m/z* calcd for $C_{19}H_{26}O_4$ (M^+): 318.1831; found: 318.1832.

(*S*)-2,2-Dimethyl-5-[1-methyl-1-(2-naphthyl)propyl]-1,3-dioxane-4,6-dione (2k)

Yield: 82%; beige solid; mp 132–134 °C; $[\alpha]_D^{31}$ –4.8 (*c* 2.7, CH_2Cl_2).

An enantiomeric excess of 79% *S* was measured by chiral HPLC [AD-H, 1% *i*-PrOH–hexanes with 0.1% TFA, 1.0 mL/min, t_{R1} = 26.0 min (*S*), t_{R2} = 39.1 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.81–7.78 (m, 3 H), 7.70 (s, 1 H), 7.46–7.41 (m, 3 H), 3.71 (s, 1 H), 2.29 (dq, *J* = 14.2, 7.4 Hz, 1 H), 2.17 (dq, *J* = 14.1, 7.2 Hz, 1 H), 1.72 (s, 3 H), 1.59 (s, 3 H), 1.14 (s, 3 H), 0.75 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 125 MHz): δ = 164.7, 164.1, 139.7, 133.1, 132.1, 128.2, 127.9, 127.3, 126.1 (2C), 126.0, 124.6, 105.2, 57.2, 46.5, 32.8, 29.4, 27.2, 21.7, 8.8.

HRMS (EI): *m/z* calcd for $C_{20}H_{22}O_4$ (M^+): 326.1518; found: 326.1519.

(*S*)-2,2-Dimethyl-5-[2-(1-tosyl-1*H*-pyrrol-3-yl)butan-2-yl]-1,3-dioxane-4,6-dione (2l)

Yield: 93%; clear oil; $[\alpha]_D^{30.5}$ +18.4 (*c* 1.5, CH_2Cl_2).

An enantiomeric excess of 97% *S* was measured by chiral HPLC [AD-H, 10% *i*-PrOH–hexanes, 1.0 mL/min, t_{R1} = 16.7 min (*S*), t_{R2} = 18.2 min (*R*)].

^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.69 (d, *J* = 8.3 Hz, 2 H), 7.26 (d, *J* = 8.2 Hz, 2 H), 7.05–7.03 (m, 1 H), 6.96 (t, *J* = 1.9 Hz, 1 H), 6.16 (dd, *J* = 3.2, 1.8 Hz, 1 H), 3.37 (s, 1 H), 2.37 (s, 3 H), 2.01 (dq, *J* = 14.0, 7.2 Hz, 1 H), 1.85 (dq, *J* = 14.1, 7.2 Hz, 1 H), 1.52 (s, 3 H), 1.44 (s, 3 H), 1.01 (s, 3 H), 0.80 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.6, 164.2, 144.9, 136.0, 131.5, 129.9, 126.8, 120.8, 118.6, 113.1, 105.2, 55.9, 42.6, 32.8, 29.7, 26.8, 22.9, 21.5, 8.6.

HRMS (EI): *m/z* calcd for $C_{21}H_{25}NO_6S$ (M^+): 419.1403; found: 419.1401.

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